Silent Voice Bridge: The Edge AI ASL Interpreter

Report 1: Problem, Competitors, and Your Plan

CMPE 195A Senior Design Project I Spring 2026

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1. Problem Statement

1.1 Problem Description

The communication barriers between DHH and hearing people is one of the main problems that need to be resolved in daily communication. DHH people mainly use American Sign Language (ASL) for communication, and ASL mainly involves hand gestures, facial movements, and movement based communication. However, the majority of the hearing population does not understand ASL, and hence, communication barriers are created in public services, work environments, and educational environments. DHH people are then forced to use interpreters and text based communication, but that is not always possible and convenient.

The main causes of the problem are the lack of ASL interpreters, the lack of real time translation services, and technological limitations in recognizing sign language gestures. The current solutions, such as interpreters, smart gloves, and cloud based sign language recognition, are limited in their own ways. The main people involved in the problem are DHH people, who use ASL for communication, and hearing people, who want to communicate with DHH people but are unable to understand ASL. The solution to the problem will definitely be beneficial in improving accessibility and inclusiveness in communication.

1.2 Problem Significance

Communication barriers may have a significant effect on the social and economic opportunities that are available to Deaf and Hard of Hearing people. According to the National Deaf Center on Postsecondary Outcomes, the employment rate of deaf people is significantly lower compared to hearing people, partly due to communication barriers in the workplace and employment [1]. Such challenges are not limited to employment; they affect various other areas of life.

Current research indicates that sign language recognition technology has the ability to overcome communication barriers by facilitating real time translation from sign language to spoken and written words [2]. Unfortunately, current technology that is available in the market heavily relies on cloud computing, wearable devices, and infrastructure, which may limit accessibility. If this problem is not solved, DHH people may still face challenges in communication, limiting social inclusion and equal access to opportunities. Developing a real time, accessible, and privacy preserving sign language translation system can help overcome these barriers and foster inclusive communication between DHH and hearing people.

1.3 Scope and Constraints

The scope of this project is to build a real-time bidirectional communication system that translates American Sign Language gestures to English text and spoken English to text for communication among Deaf and Hard of Hearing people. The system will employ a webcam to capture sign language gestures and an NVIDIA Jetson edge computing system to recognize these gestures in real time using computer vision techniques. The text will be displayed in a dynamic web interface for communication purposes.

This project will specifically focus on recognizing the ASL alphabet, a number of static signs, and a limited number of common phrases used in sign language. Advanced linguistic complexities in ASL, such as complex grammar rules in sign language, facial expressions, and fluent conversations, are beyond the scope of this project. The system will also not focus on scalability performance; rather, it will emphasize proof of concept performance.

Technical Constraints: Hardware limitations: The system would depend on the computational power provided by the NVIDIA Jetson and the average 1080p webcam.

Software/platform requirements: The system would need to incorporate various technologies such as Python, OpenCV, MediaPipe, TensorFlow, PyTorch, and a React frontend.

Technical dependencies: The system would depend on the accurate calculation of hand and pose landmarks using MediaPipe and efficient models that can run in real time on edge devices.

Non-Technical Constraints: Time constraints: The project needs to be finished within the two-semester senior design schedule, which includes CMPE 195A and CMPE 195B.

Budget/resource constraints: In order to keep the costs low, the hardware used will be affordable and readily available, such as the NVIDIA Jetson platform and webcams.

Regulatory or compliance constraints: The system should take into consideration the best practices regarding accessibility and the privacy of the user by processing the information locally without sending any video data to the server.

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2. Competitor Analysis

2.1 Existing Solutions

[Identify at least 3 existing solutions or approaches to this problem. For each, describe what it does, its strengths and weaknesses, and its target audience.]

Competitor 1: SignAll

SignAll is a kiosk-based ASL translation system that uses multiple cameras and sensors to capture sign language and translate it to text or speech. Its key strength is high accuracy due to its multi-camera setup and controlled environment. However, it's expensive, not portable, and requires a dedicated installation space. Its target audience is institutions like government offices, hospitals, and universities that can afford permanent installations.

Competitor 2: Google Live Transcribe / Ava

These are smartphone apps that provide real-time speech-to-text transcription. They're free or low-cost, widely accessible, and work on devices people already own. However, they don't translate ASL. They only transcribe spoken language, which doesn't help deaf individuals who communicate primarily through sign language. Their target audience is people with hearing loss who can still read lips or use spoken language.

Competitor 3: Hand Talk

Hand Talk is a mobile app that translates text and speech into sign language using a 3D avatar. It helps hearing people communicate with deaf individuals by showing them signs. However, it works in only one direction. It doesn't translate ASL into text for deaf individuals to communicate outward. Its target audience is hearing people who want to learn or use sign language, not deaf individuals seeking independent communication.

2.2 Differentiation Opportunity

Current solutions fail to address the core problem: giving deaf individuals the ability to communicate independently in any environment. SignAll requires expensive, fixed installations that aren't practical for everyday use. Apps like Google Live Transcribe don't support ASL at all as they assume the user can speak or read lips. Hand Talk only translates in one direction, helping hearing people understand sign language but not helping deaf individuals express themselves.

Our ASL translation device fills this gap by providing real time ASL to text translation in a portable, offline device. Unlike SignAll, it's affordable and can be used anywhere. Unlike transcription apps, it actually recognizes ASL signs. Unlike Hand Talk, it empowers deaf individuals to communicate outward rather than relying on others to learn sign language. By running entirely offline on edge

hardware, we also address privacy concerns that cloud-based solutions raise, as no video data is ever sent to external servers.

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3. Your Proposed Solution

3.1 Solution Overview We propose a real time, bidirectional ASL to English translation system. The hardware consists of a standard webcam connected to an NVIDIA Jetson edge computing device. As the DHH user signs, the camera captures the video feed, and the Jetson locally processes the footage using computer vision and machine learning to classify the gestures. The translated English text is then instantly displayed on a responsive web interface built with React, which the hearing user can read. The system will also feature a speech to text component, allowing the hearing user to speak and have their words displayed on the screen for the DHH user.

3.2 Technical Approach

- **Technologies & Tools:**
    1. **Frontend:** React, JavaScript, HTML/CSS (for a responsive, accessible UI).
    2. **Backend / AI Logic:** Python, OpenCV (image processing), Google MediaPipe (hand and pose landmark extraction), TensorFlow/PyTorch (gesture classification model).
    3. **Hardware:** NVIDIA Jetson Nano/Orin (for hardware accelerated AI inference), standard 1080p webcam.
- **Core Technical Challenges:**
    1. **Challenge 1: Real time Latency.** Processing video frames through neural networks can cause lag. *Solution:* We will use MediaPipe to extract lightweight skeletal landmark coordinates (x, y, z points) instead of feeding raw image pixels into the classification model, drastically reducing the computational load on the Jetson.
    2. **Challenge 2: Dynamic Sign Translation.** ASL relies heavily on motion over time, not just static hand shapes. *Solution:* We will utilize an LSTM (Long Short Term Memory) or Transformer based neural network architecture capable of analyzing sequences of frames to understand motion context.
    3. **Challenge 3: Background Noise and Lighting.** Cluttered backgrounds can confuse computer vision models. *Solution:* MediaPipe's robust hand tracking backend naturally isolates hands from backgrounds, but we will also implement preprocessing steps in OpenCV to normalize lighting and contrast before landmark extraction.

3.3 Value Proposition Unlike smart gloves, our solution is entirely non intrusive, the user only needs to stand in front of a camera. Unlike cloud based AI apps, our solution uses the NVIDIA Jetson to process data on the edge, which guarantees low latency, eliminates reliance on Wi Fi, and ensures absolute data privacy since no video feeds are ever uploaded to the internet. This creates a secure, fast, and natural communication environment.

3.4 Success Criteria

- **CMPE 195A Goals (Proof of Concept):**
    1. Successfully deploy the MediaPipe hand tracking pipeline on the NVIDIA Jetson and maintain a minimum of 30 FPS.
    2. Train a baseline machine learning model capable of accurately translating the static ASL alphabet (A Z) and 5 10 static words.
    3. Establish the communication pipeline between the Python backend and the React frontend, displaying translated text on the UI with under 300ms of latency.

- **CMPE 195B Goals (Full Implementation):**
    1. Expand the model to translate dynamic signs (words requiring movement) using sequence based neural networks (e.g., LSTMs).
    2. Implement bidirectional communication, integrating a speech to text API so hearing users can speak back to the DHH user via the React UI.
    3. Achieve an overall translation accuracy of 85% or higher on a curated dictionary of 50+ common conversational ASL phrases in varied lighting conditions.

References

[1] C. Garberoglio, S. Palmer, S. Cawthon, and R. Sales, "Deaf People and Employment in the United States: 2019," National Deaf Center on Postsecondary Outcomes, Austin, TX, 2019.

[2] D. Bragg et al., "Sign Language Recognition, Generation, and Translation: An Interdisciplinary Perspective," in The 21st International ACM SIGACCESS Conference on Computers and Accessibility, Pittsburgh, PA, USA, 2019, pp. 16–31.

[3] C. Lugaresi et al., "MediaPipe: A Framework for Building Perception Pipelines," arXiv preprint arXiv:1906.08172, 2019. [Online]. Available: https://arxiv.org/abs/1906.08172. [Accessed: March 13, 2026].